

plotly_matlab: interactive web-based graphing for GNU Octave and MATLAB

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Software

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Summary

plotly_matlab is the official Plotly graphing library for MATLAB and GNU Octave. It converts native figures, created with either engine, into interactive, web-based charts that can be explored, exported, and embedded in web pages with a single line of code.

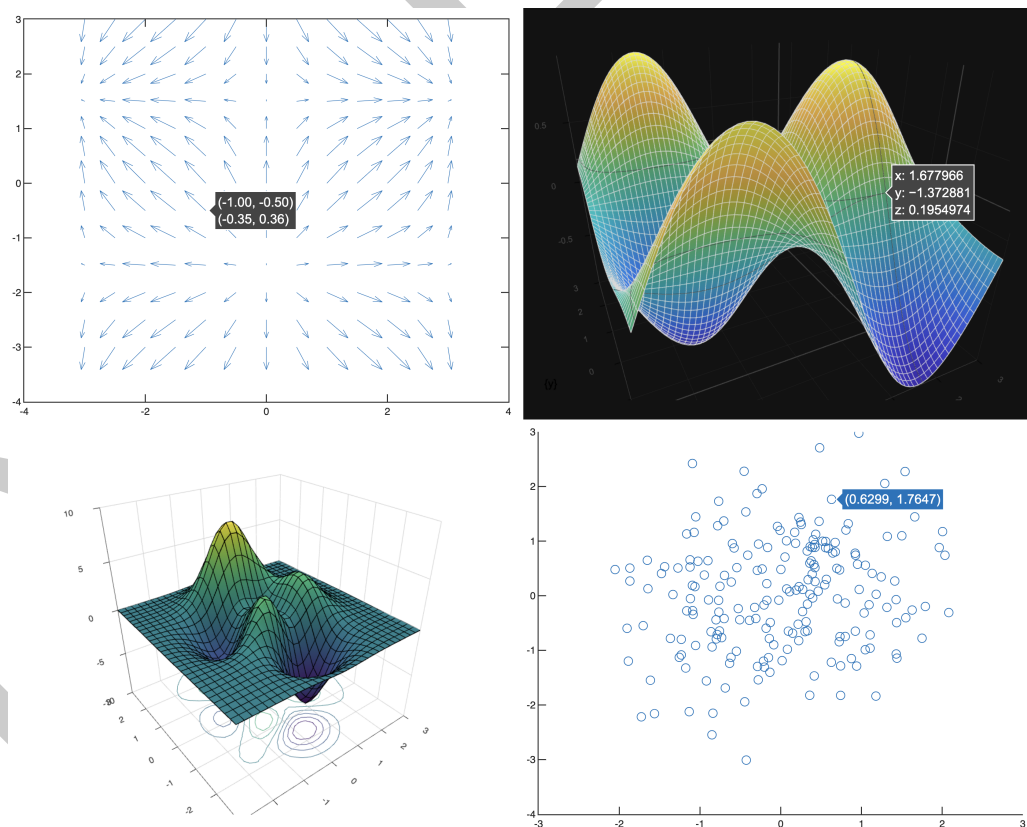


Figure 1: Example plots generated by this library in Octave (top-left: quiver; top-right: ezsurf; bottom-left: surf; bottom-right: scatter).

Interactive web-based plots play a central role in scientific research. Because the charts are interactive, viewers can zoom and pan to inspect fine detail, and rotate three-dimensional charts to change perspective. Hovering over data points displays tooltips with numerical values. Because the charts are web-based, they are easy to share between researchers on different operating systems, unlike language-specific figure objects.

15 First released in 2013, this library brings interactive web-based plots to Octave and MATLAB.

16 Statement of need

17 Plotly ([Sievert, 2020](#)) is widely used by the scientific community to generate interactive web
18 plots (its canonical scholarly citation has been cited more than 1,000 times on OpenAlex¹).
19 While other languages were supported by Plotly, the Octave ([Eaton et al., 2023](#)) and MATLAB
20 ([Higham & Higham, 2016](#)) ecosystems, which are widely used in scientific research, both
21 lacked a plotting library that is interactive and web-based. This library bridges this gap by
22 providing an easy way to generate Plotly figures from a broad range of native plots supported
23 by Octave and MATLAB.

24 State of the field

25 Wrappers for Plotly also exist in R ("[Plotly.R](#)," n.d.), Python ("[Plotly.py](#)," n.d.), Rust
26 ("[Plotly.rs](#)," n.d.), and .NET programming languages ("[Plotly.NET](#)," n.d.). For MATLAB and
27 Octave, the existing alternatives cover only part of the workflow: `export_fig` ([Altman, n.d.](#))
28 is a widely used tool for high-quality static export of figures but produces no interactive charts;
29 `matlab2tikz` ("[matlab2tikz](#)," n.d.) targets LaTeX documents with generated TikZ code;
30 and the `gramm` package ([Morel, 2018](#)) provides a grammar-of-graphics interface for statistical
31 visualization in MATLAB but renders to native figures rather than web charts. Rather than
32 extending an existing binding, this library implements figure conversion directly from each
33 engine's native graphics object model, providing Octave and MATLAB users with a dedicated
34 workflow for interactive web-based graphing.

35 Software design

36 The core of the library is the `plotlyfig` class, which walks the graphics object tree of a figure
37 (figure, axes, and children) and builds a Plotly figure description (data and layout) through
38 per-class dispatch: a central `updateData` function routes each graphics object to a dedicated
39 `updateX` function (one per supported plot type), with shared geometry and color helpers
40 used across plot types. `hgggroup`-based plot types are resolved through engine-aware group
41 classification: both engines wrap plot types in `hgggroups` (groups of primitive graphics objects
42 such as lines, patches, and text), but the group structures differ, so the wrapper identifies the
43 plot type behind each group before dispatch. Three-dimensional scenes are handled explicitly,
44 including camera geometry and axis aspect-ratio computation derived from each engine's native
45 camera model.

46 A central design commitment is cross-engine parity. MATLAB and GNU Octave differ in
47 their graphics object trees (e.g., Octave wraps plot types in `hgggroups` that MATLAB exposes
48 differently) and property semantics (character arrays versus string types, annotation properties
49 absent in Octave). The library maintains a single codebase that provides consistent behavior
50 across both engines, with behavior verified by a regression suite that runs on each.

51 Testing is an essential part of the project. A test runner (`runplotlytests`) executes structural
52 regression tests of converted figure descriptions in both engines, and a gallery of 81 plot types
53 (`makegallery`) generates native and converted outputs that can be compared by rendering
54 with Plotly.js in a browser. Continuous integration runs the Octave test suite on every push
55 (GitHub Actions).

¹Cited 1,061 times as of Aug 2026 (<https://openalex.org/works/W2991157178>)

Research impact statement

plotly_matlab is used in published scientific research. For example, it was used to produce visualizations in a biomechanical imaging study of pelvic floor ligaments (Matter et al., 2024). The library is distributed through the official Plotly MATLAB page (plotly.com/matlab) and the MathWorks File Exchange. The library is currently maintained and supported through the Plotly community forum. Its public GitHub history documents tagged releases and extensive issue triage.

AI usage disclosure

Gemini 3.1 Pro was used exclusively as assistance in the writing of this manuscript for structure and clarity. All AI suggestions were reviewed and validated by the authors. The authors made all decisions and verified technical accuracy.

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